

Fast Transient Analysis of Power Distribution Networks via Equivalent Circuit Modeling

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Abstract— *This paper presents an efficient method to analyze power distribution networks in the time-domain. Instead of directly analyzing the integration approximated power/ground networks at each time step as previous methods did, the new method first builds the equivalent models for many series RLC-current chains based on their Norton's form companion models in the original networks, and then the Incomplete-Cholesky Conjugate Gradient (ICCG) based iterative method is used to solve the reduced networks which is symmetric positive definite. The solutions of the original networks then are back solved from that of the reduced networks. The success of the new algorithm simply lies on the exploitation of the regularities in the power/ground networks. Experimental results demonstrated that the new algorithm is one order of magnitude faster than the pure ICCG method and two order of magnitude faster than SPICE at almost no accuracy loss. The new algorithm takes minutes to simulate a P/G network in time domain with more than 1 million nodes on common workstations*

I. INTRODUCTION

Signal integrity in the power/ground (P/G) bus is emerging as a limiting factor in the nano-regime VLSI chip designs. As P/G grids experiences the largest current flows in a chip and are more susceptible to current-induced reliability and functional failures. Efficient transient analysis techniques of P/G grids are required to precisely capture the dynamic voltage fluctuations on the P/G wires to guide the designs of a reliable and working P/G networks. As millions of devices are integrated into a single chip, transient analysis of a power/ground network become a challenging task because of the increasing sizes of the underlying P/G networks. The traditional circuit simulators method like SPICE is no longer able to meet the formidable tasks of analyzing P/G circuits with millions of extracted RLKC elements in a timely manner.

Due to the increasing importance of P/G integrity issues such as excessive IR drops, Ldi/dt noise, electromigration and resonance, significant research efforts have been carried out to find efficient simulation approaches to P/G grid analysis [2, 5, 3, 4, 6, 8, 9, 10] in the past. Among them, multigrid method [8], hierarchical method [10], preconditioned conjugate gradient (PCG) [3], hierarchical model order reduction [4]

are the latest methods proposed. Although significant improvements have been made to analyze large P/G grids by those proposed methods, the naturally regular physical structures of a P/G network is not fully explored to speedup transient analysis of networks.

In this paper, we propose a new approach to the transient analysis of large P/G grids. Our new method is based on the observation that P/G grids, especially those used in many ASIC applications, consist of any series RLC chains as shown in Fig. 1. We show that a compact equivalent model can be built for the RLC chain circuit in each time step of approximate integration. With the equivalent chain models, the original networks can be reduced significantly and efficiency transient analysis of very large P/G networks become possible. As only resistors and constant current sources are present in the reduced P/G networks in each time step, the resulting circuit matrix of the network is symmetric positive definite (the voltage sources are also modeled by Norton equivalent circuits), preconditioned conjugate gradient based iterative method [7] is used to efficiently solve those circuit matrices. We notice that similar idea was proposed by Stark and Horowitz [9] for DC analysis of resistor-only P/G grids.

This paper is organized as follows. Section II presents how equivalent circuit models are constructed for RLC chain circuit in P/G networks. Section III gives the general flows of the proposed algorithm. Experimental results are described in Section IV. Section V concludes the paper.

II. CONSTRUCTION OF EQUIVALENT CIRCUIT MODELS

The P/G networks are typically mesh-structured in today's VLSI technology and consist of many cascaded RLC sections extracted from chip layouts as shown in Fig. 1 for a power network.

Here R_i, C_i, L_i denote, respectively, the resistance, capacitance, and inductance at P/G wire segment i . Capacitance C_i include both the parasitic capacitances of the P/G wire i and the decoupling capacitances around. e_i is the time-varying current source that captures the dynamic current consumptions of circuit macros. N_1 and N_{n+1} are called *cross nodes* and the rest of the nodes are called *intermediate nodes*. Our goal is to suppress all the intermediate nodes in this series RLC chain circuit

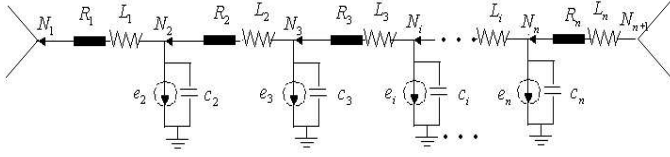


Fig. 1: A series RLC chain in a P/G network.

and replace them by an electrically equivalent circuit that consists of only the cross nodes in each integration time step of transient analysis.

A. Integration Approximation In Norton's Form

Let h be the time step used at simulation step $k + 1$. For capacitors and inductors, trapezoidal companion models of Norton's form are used. In this way, we will not introduce any extra nodes in the circuit matrix. Specifically, the current through a capacitor at $k + 1$ step is:

$$I_{C,k+1} = \frac{2C}{h}V_{C,k+1} - \left(\frac{2C}{h}V_{C,k} + I_{C,k}\right), \quad (1)$$

where $V_{C,k}, I_{C,k}, V_{C,k+1}, I_{C,k+1}$ denote, respectively, the branch voltages and branch currents of the capacitor at step k and at step $k + 1$ respectively and C is the value of the capacitor. Similarly the current through an inductor at step $k + 1$ is:

$$I_{L,k+1} = \frac{h}{2L}V_{L,k+1} + \left(\frac{h}{2L}V_{L,k} + I_{L,k}\right), \quad (2)$$

where $V_{L,k}, I_{L,k}, V_{L,k+1}, I_{L,k+1}$ denote, respectively, the branch voltages and branch currents of the inductor at step k and at step $k + 1$ and L is the value of the inductor. The trapezoidal Norton companion models for a capacitor and an inductor are shown in Fig. 2 and Fig. 3.

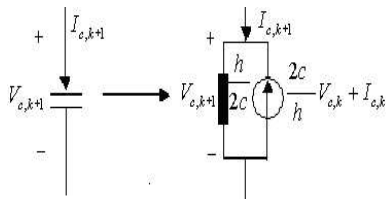


Fig. 2: Trapezoidal companion model for a capacitor.

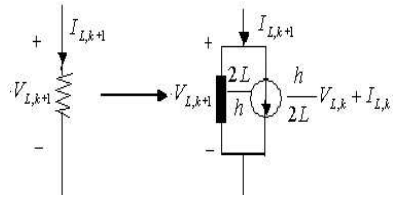


Fig. 3: Trapezoidal companion model for an inductor.

Let's go back the series RLC chain and mark each capacitor and inductor with the integration time step index shown in Fig. 4 at $k + 1$ step, the RLC chain between two cross nodes V_1 and V_{n+1} become Fig.5 when all the capacitors and inductors are replaced by their companion models at time step $k + 1$ shown in Fig.4, $E_{i,k+1}$ denotes the current flowing into node i from node $i + 1$ on step $k + 1$.

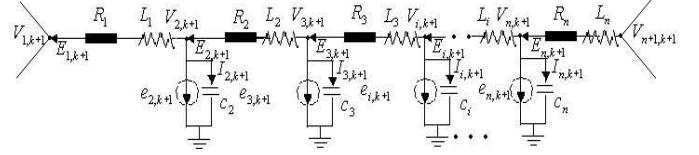


Fig. 4: The RLC chain circuit at time step $k + 1$.

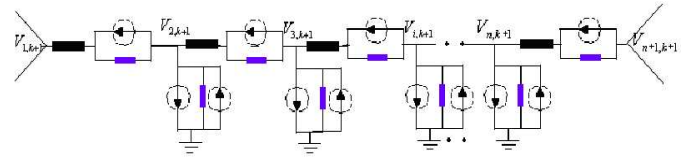


Fig. 5: The RLC chain circuit in terms of companion models at time step $k + 1$.

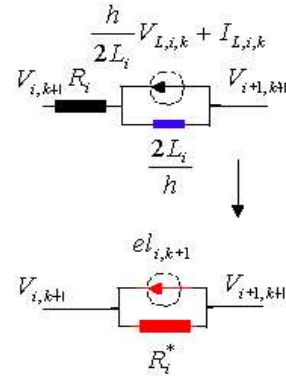


Fig. 6: Merge two resistors in a RLC section.

Next, we merge two floating resistors for each RLC section as shown in Fig. 6 using Norton theory. Then we have

$$R_i^* = 2L_i/h + R_i, \quad (3)$$

$$el_{i,k+1} = \left(\frac{h}{2L_i}V_{L,i,k} + I_{L,i,k}\right) \cdot \frac{2L_i/h}{2L_i/h + R_i}, \quad (4)$$

where $V_{L,i,k}$ and $I_{L,i,k}$ are the branch voltage and branch current of L_i at step k . They can be represented by node voltages and branch currents:

$$I_{L,i,k} = E_{i,k}$$

$$V_{L,i,k} = (V_{i+1,k} - V_{i,k}) - E_{i,k}R_i.$$

We then combine two grounded current sources, one is coming from the independent current source and another is coming from capacitor's companion model, into one current source $ec_{i,k+1}$ for each RLC section:

$$ec_{i,k+1} = e_{i,k+1} - \left(\frac{2C_i}{h} V_{i,k} + I_{i,k} \right), \quad (5)$$

and rename the resistor from C_i 's companion model as $r_i = \frac{h}{2C_i}$. So the circuit in Fig. 5 can be simplified to the following circuit shown in Fig. 7, which is more ready for further reduction shown in the next subsection.

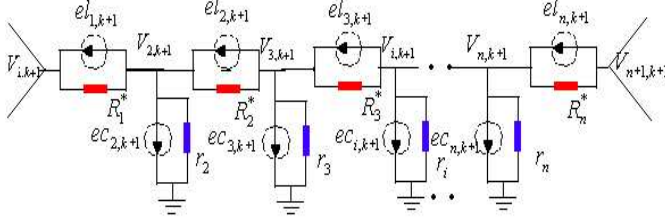


Fig. 7: Node-reduced RLC chain circuit.

B. Transformation From $\mathbf{Y}$ Model to π Model

In this subsection, we show how RLC chain circuit shown in Fig 7 can be further reduced by an equivalent circuit consisting of only the cross nodes. This is done by repeatedly transferring a $\mathbf{Y}$ model circuit to a π model circuit as shown in Fig. 8:

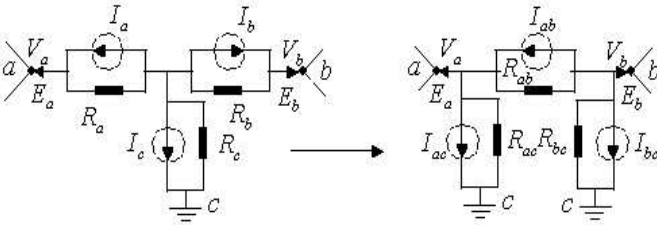


Fig. 8: $\mathbf{Y}$ model circuit to π model circuit.

In Fig. 8, E_a and E_b are the currents inflowing into node a and node b respectively, and the corresponding arrowheads show the current directions. To compute the equivalent currents and resistances shown in the right-hand side of Fig. 8, we first look at node a and we have

$$\begin{aligned} V_b - V_a &= R_a(E_a - I_a) \\ &\quad + R_b(E_a + I_c + \frac{V_a + R_a(E_a - I_a)}{R_c} + I_b) \\ &= \left(R_a + R_b + \frac{R_a R_b}{R_c} \right) E_a + R_b I_c + R_b I_b \\ &\quad + \frac{R_b}{R_c} V_a - R_a I_a - \frac{R_a R_b}{R_c} I_a. \end{aligned}$$

We define the following equivalent resistors:

$$R_{ab} = \frac{R_a R_b + R_a R_c + R_b R_c}{R_c}, \quad (6)$$

$$R_{ac} = \frac{R_a R_b + R_a R_c + R_b R_c}{R_b}, \quad (7)$$

$$R_{bc} = \frac{R_a R_b + R_a R_c + R_b R_c}{R_a}. \quad (8)$$

As a result, we have

$$E_a = \frac{V_b - V_a}{R_{ab}} - \frac{V_a}{R_{ac}} + \left(\frac{R_a I_a}{R_{ab}} - \frac{R_b I_b}{R_{ab}} \right) - \left(\frac{R_c I_c}{R_{ac}} - \frac{R_a I_a}{R_{ac}} \right)$$

Similar result can be obtained for node b :

$$E_b = \frac{V_a - V_b}{R_{ab}} - \frac{V_b}{R_{bc}} - \left(\frac{R_a I_b}{R_{ab}} - \frac{R_b I_b}{R_{ab}} \right) - \left(\frac{R_c I_c}{R_{bc}} - \frac{R_b I_b}{R_{bc}} \right)$$

We further define the following equivalent currents:

$$I_{ab} = \frac{R_a I_a}{R_{ab}} - \frac{R_b I_b}{R_{ab}}, \quad (9)$$

$$I_{ac} = \frac{R_c I_c}{R_{ac}} - \frac{R_a I_a}{R_{ac}}, \quad (10)$$

$$I_{bc} = \frac{R_c I_c}{R_{bc}} - \frac{R_b I_b}{R_{bc}}. \quad (11)$$

Finally we can represent E_a and E_b in terms of those equivalent currents and resistances as follows:

$$E_a = \frac{V_b - V_a}{R_{ab}} - \frac{V_a}{R_{ac}} + I_{ab} - I_{ac}, \quad (12)$$

$$E_b = \frac{V_a - V_b}{R_{ab}} - \frac{V_b}{R_{bc}} - I_{ab} - I_{bc}. \quad (13)$$

Equations (12) and (13) essentially gives us the π model representation of the same circuit as shown in the right-hand side of the Fig. 8.

C. Constructions and Back Solving Algorithms for the Equivalent Circuit Models.

By repeatedly applying the $\mathbf{Y}$ model to π model transformation starting from node N_1 until we reach node N_{n+1} , the whole chain circuit in Fig. 7 can be reduced into an equivalent circuit shown in Fig. 9:

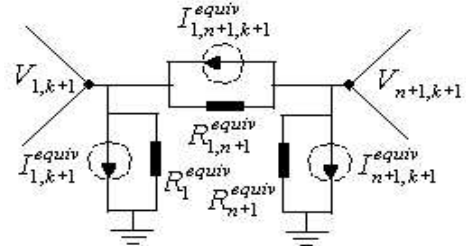


Fig. 9: The final reduced equivalent chain circuit model.

The algorithm for computing R_1^{equiv} , R_{n+1}^{equiv} , $R_{1,n+1}^{equiv}$, $I_{1,k+1}^{equiv}$, and $I_{n+1,k+1}^{equiv}$ from the circuit in Fig. 7 is shown in Fig.10.

Construct Equivalent Circuit Model

EQVCKT-CONSTRUCT()

```
{
   $R_1^{equiv} = \infty, I_{1,k+1}^{equiv} = 0, r_{n+1} = \infty, ec_{n+1,k+1} = 0.$ 
   $R_a = R_1^*, R_c = r_2, I_a = el_{1,k+1}, I_c = ec_{2,k+1}.$ 
   $j = 2$ 
  while  $j \leq n$  do
  {
     $R_b = R_j^*, I_b = -el_{j,k+1}.$ 
    compute  $R_{ab}, R_{ac},$  and  $R_{bc}$  based on (6), (7) and (8).
    compute  $I_{ab}, I_{ac}$  and  $I_{bc}$  based on (9), (10) and (11).
     $j = j + 1$ 
     $R_a = R_{ab}, R_c = R_{bc} || r_j, R_1^{equiv} = R_1^{equiv} || R_{ac}.$ 
     $I_a = I_{ab}, I_c = I_{bc} + ec_{j,k+1}, I_{1,k+1}^{equiv} = I_{1,k+1}^{equiv} + I_{ac}.$ 
  }
   $R_{n+1}^{equiv} = R_c, I_{n+1,k+1}^{equiv} = I_c, R_{1,n+1}^{equiv} = R_a, I_{1,n+1,k+1}^{equiv} = I_a.$ 
}
```

Fig. 10: Algorithm for constructing equivalent circuit model.

Once the reduced network has been solved, all the intermediate nodes of original circuits can be back solved using the superposition principle. The algorithm for computing all the node voltages of intermediate nodes $V_{j,k+1}$ is shown in Fig. 11.

Back Solve the Intermediate Node Voltages

BACK-SOLVER()

```
{
   $E_{1,k+1} = \frac{V_{n+1,k+1} - V_{1,k+1}}{R_{1,n+1}^{equiv}} + I_{1,n+1,k+1}^{equiv} - I_{1,k+1}^{equiv} - \frac{V_{1,k+1}}{R_1^{equiv}}.$ 
  for  $j = 2$  to  $n$  do
  {
     $V_{j,k+1} = V_{j-1,k+1} + R_{j-1}^* (E_{j-1,k+1} - el_{j-1,k+1}).$ 
     $E_{j,k+1} = E_{j-1,k+1} + \frac{V_{j,k+1}}{r_j} + ec_{j,k+1}.$ 
  }
}
```

Fig. 11: Algorithm for back solving intermediate node voltages.

III. OVERALL ALGORITHM

After the reduced network is obtained at each time step, iterative method is used as the resulting circuit matrix is symmetric positive definite and the circuit consists of only resistors and constant current sources. Incomplete Cholesky Conjugate Gradient (ICCG) method [7] is used to solve resulting linear equations.

Assume initial node voltages are known and the number of RLC sections in a P/G network is N . The overall simulation algorithm is shown in Fig. 12.

n_{step} is the number of time steps used in the simulation. RENEW($ec_{i,k}, el_{i,k}$) is to compute the combined current sources from previous simulation results according to (4) and (5). EQVCKT-CONSTRUCT() constructs the equivalent circuits for

PG-SOLVER()

```
{
  Initiate  $V_i, R_i^*, r_i$  for  $i \in [1, N].$ 
  for  $k = 1$  to  $n_{step}$ 
  {
    RENEW( $ec_{i,k}, el_{i,k}$ )
    EQVCKT-CONSTRUCT ()
    ICCG-SOLVER ()
    BACK-SOLVER ()
  }
}
```

Fig. 12: Overall simulation algorithm.

all the RLC chains as shown in Fig.10. ICCG-SOLVER() is the linear solver based on ICCG algorithm to solve the reduced the network. BACK-SOLVER () is used to back solve all node voltages of all intermediate nodes shown in Fig. 12.

IV. EXPERIMENTAL RESULTS

The proposed simulation algorithm is implemented in C and C++. All the experimental results are obtained on SUN Ultra Sparc workstation with 250MHz CPU and 256MB memory. The number of time steps, n_{step} is 120 for a clock cycle.

We tested our program on a number of P/G networks with complexities ranging from 2000 nodes to 1 million nodes. The statistics of those P/G circuits are shown in Table I.

First, we compare our method with that of SPICE in terms of accuracy. Both programs are used to solve a 50×50 P/G network with 5 P/G strips. The results are shown in Fig. 13. The figure in the left-hand side is a node voltage waveform from the new method and the figure in the right-hand side is the same node voltage waveform from SPICE. The two waveforms are essentially the same.

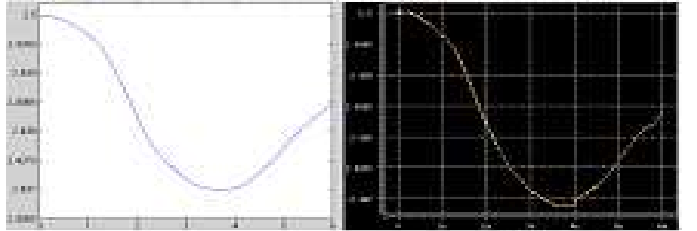


Fig. 13: Accuracy comparison with SPICE

Next, we compare our new simulation method with the SPICE and the pure ICCG algorithm (no node reduction is performed). The effectiveness of equivalent circuit modeling is shown in Table. I where the number of nodes after node reduction and the reduction ratio for each circuit are shown in column 3 and 4. It can be seen that complexities of the P/G network can be significantly reduced. On the other hand, we notice that the topologies of P/G grids do have impacts on the performances of the new algorithm.

Table I and Table II show that the pure ICCG can't deal with much large P/G networks. This is because ICCG will become

TABLE I
STATISTICS ON THE ORIGINAL AND REDUCED NETWORKS

P/G Grids	Nodes seen by ICCG		Ratio
	Original	Reduced	
25*100	2,526	126	20
25*400	10,026	126	80
25*800	20,026	126	159
50*800	40,051	251	160
200*800	160,201	1,001	160
400*900	360,401	2,001	180
500*1k	500,501	25,01	200
1k*1k	1,001,001	5,001	200

TABLE II
COMPARISON RESULTS ON CPU TIME

P/G Grids	CPU time (sec.)			Speedup over	
	SPICE	Original	Reduced	SPICE	ICCG
25*100	47.54	10.53	0.84	56.60	12.54
25*400	842.16	64.67	2.78	302.94	23.26
25*800	2397.26	147.35	5.4	443.94	27.29
50*800	N/A	320.39	12.91	N/A	24.82
200*800	N/A	2,003.44	97.44	N/A	20.56
400*900	N/A	6,029.15	225.82	N/A	26.70
500*1k	N/A	N/A	320.87	N/A	N/A
1k*1k	N/A	N/A	642.5	N/A	N/A

memory hungry when large circuits are loaded. On the other hand, the new algorithm is much more memory efficient and powerful than the pure ICCG method.

From Table II, it appears that the new algorithm is one order of magnitude faster than the pure preconditioned conjugate gradient method (ICCG) and two order of magnitudes (56X-443X) faster than SPICE. The speedup of the new algorithm is comparable with the recently proposed P/G simulation algorithm in [4].

The capability of the new algorithm is also significantly improved compared with the pure ICCG as shown in Table II. The maximum P/G networks can be solved by the ICCG method is 400 * 900 with 160k nodes and it takes 6029.15 seconds to solve the circuits. While the new algorithm can easily solve 1000*1000 P/G network in 642.5 seconds. This compares very favorably with a similar preconditioned conjugate gradient method proposed in [3] where the maximum P/G network with 260,820 nodes is solved in 2150 seconds in time domain. We notice that our CPU and memory configuration are far from that in the state of the art workstations and our time step is sufficient small.

V. CONCLUSION

This paper proposes an efficient technique for analysis of power distribution networks of deep sub-micron VLSI systems in time-domain. At each time step, after integration approximation by Norton-form companion models, equivalent circuit models are built for many RLC chain circuits to reduce the P/G network complexities. Then Incomplete-Cholesky Conjugate Gradient (ICCG) based iterative method is used to solve the reduced resistor-current only networks. Experimental results demonstrated that the new algorithm is one order of magnitude faster than the pure ICCG method and two order of magnitude faster than SPICE at almost no accuracy loss. The new algorithm takes 642.5 seconds to solve a P/G networks with more than 1 million nodes on a 250Mhz Sun workstation. This makes our algorithm a serious contender for attacking real industry P/G networks.

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